Theory Radar: A* Formula Search with a Provably Admissible DAG Heuristic

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***Abstract*—We present Theory Radar, an A* search algorithm for discovering interpretable symbolic classifiers from labeled data. The search operates over formula trees with cost equal to formula depth, using a heuristic $h = \max(h_1, h_2, h_3)$ composed of three individually admissible components: a target-check heuristic, an AUROC-F1 bound derived from a proven tight inequality, and a feature-coverage heuristic. In *strict mode*, the algorithm provides a provable optimality guarantee: the first formula meeting a specified F1 target is the shallowest such formula in the search space. In *fast mode*, an optional AUROC subtree screening heuristic reduces evaluations by 70–99% with no observed admissibility violations across 56 test conditions, at the cost of losing the formal guarantee.**

Three supporting theorems are proved. The *Monotone Invariance Theorem* shows that strictly monotone transforms preserve the optimal thresholded F1 score, allowing the search to skip redundant F1 computations while still expanding descendants. The *AUROC Rank-Order Invariance* theorem shows AUROC is preserved under monotone transforms. The *AUROC-F1 Bound* theorem provides a tight upper bound on the maximum achievable F1 for a given AUROC, derived from the step-function ROC curve.

Across 9 classification datasets, Theory Radar discovers formulas that phased enumeration misses: on Moons, it finds $\max(\mathtt{x2} - \mathtt{x1}, \mathtt{x2})$ (F1 $= 0.893$) versus the phased ceiling of 0.833; on Circles, the formula $\sqrt{\mathtt{x1}^2 + \mathtt{x2}^2}$ achieves F1 $= 0.997$ and statistically significantly outperforms a gradient-boosted ensemble ($p = 0.036$, 5-fold CV).

Code: https://github.com/ahb-sjsu/theory-radar (`pip install theory-radar`).

I. Introduction

Symbolic prediction seeks interpretable formulas that classify data by thresholding a mathematical expression: $\hat{y} = \mathbf{1}[f(\mathbf{x}) > \tau]$. Existing approaches include genetic symbolic regression [1], [3], neural equation discovery [2], search-based symbolic regression [4], and phased exhaustive enumeration [5]. Genetic methods explore deep formula trees but provide no optimality guarantee at a given depth. Neural methods produce opaque representations. Search-based methods enumerate small expressions systematically but typically optimize regression loss rather than classification metrics. Phased enumeration is complete within fixed templates but misses formulas with other structures (e.g., pairwise-of-pairwise compositions).

We propose **Theory Radar**, an A* search over formula trees with a provably admissible heuristic. The search minimizes formula depth subject to achieving a target F1 score. The heuristic is a DAG (directed acyclic graph) of three admissible components combined via $\max$, which is itself admissible [6].

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In *strict mode*, no subtree pruning is applied and the optimality guarantee holds. In *fast mode*, an empirical AUROC screening filter provides 70–99% speedup without observed admissibility violations.

The algorithm also computes F1 via exact threshold optimization (sorting all N samples and sweeping all possible thresholds in $O(N \log N)$), rather than the percentile approximation used in prior work. This eliminates threshold-related noise from the evaluation.

II. Theoretical Results

Throughout, we define the classifier as $\hat{y} = \mathbf{1}[f(\mathbf{x}) > \tau]$ *or* $\hat{y} = \mathbf{1}[f(\mathbf{x}) < \tau]$, with the direction chosen to maximize F1 (both are evaluated). AUROC is computed as $\max(\mathrm{AUC}, 1 - \mathrm{AUC})$ to ensure direction invariance. These conventions are used consistently throughout.

Theorem 1 (Monotone Invariance)**.** *Let g be strictly monotone on the range of f. Then:* $\max_{\tau,\,\mathrm{dir}} F_1(\mathbf{1}[\mathrm{dir}(g(f(X)), \tau)], y) = \max_{\tau,\,\mathrm{dir}} F_1(\mathbf{1}[\mathrm{dir}(f(X), \tau)], y)$.

Proof. If g is increasing, $g(f(x)) > \tau \Leftrightarrow f(x) > g^{-1}(\tau)$. If decreasing, $g(f(x)) > \tau \Leftrightarrow f(x) < g^{-1}(\tau)$. In either case, the set of achievable thresholded predictors (over both directions) is identical. □

Corollary 1 (Monotone F1 Reuse)**.** *When expanding a node f via a monotone unary operation g, the child $g(f)$ has the same optimal F1 as f. The search may reuse the parent's F1 value, saving one $O(N \log N)$ evaluation. Note: this does* not *justify pruning the subtree rooted at $g(f)$, since descendants like $(g(f))^2$ may differ from f^2.*

Theorem 2 (AUROC-F1 Bound)**.** *For* AUROC $= A$ *and prevalence π, the maximum thresholded F1 satisfies:*

$$F_1^{\max}(A, \pi) \leq \sup_{t \in [2A-1,\,1]} \frac{2t\pi}{t\pi + \pi + (t - 2A + 1)(1 - \pi)} \tag{1}$$

This bound is tight (achieved by the step-function ROC curve).

Proof. Any ROC curve with area A has Youden index $J = \max(TPR - FPR) \leq 2A - 1$. At operating point (FPR, TPR) with $FPR = TPR - J$, substitute into $F_1 = 2 \cdot TPR \cdot \pi / (TPR \cdot \pi + \pi + FPR \cdot (1 - \pi))$ and maximize over $TPR \in [J, 1]$. □

III. A* Search with DAG Heuristic

Definition 1 (DAG Heuristic)**.** $h(n) = \max(h_1(n), h_2(n), h_3(n))$ *where:*

- h_1*: 0 if* $F_1(n) \geq t$ *(target met), 1 otherwise. [*$O(1)$*]*
- h_2*: 0 if* $F_1^{\max}(\mathrm{AUROC}(n), \pi) \geq t$ *(Theorem 2), 1 otherwise. [*$O(N \log N)$*]*
- h_3*: 0 if any single feature achieves* $F_1 \geq t$ *(precomputed), 1 if the formula uses* ≤ 1 *feature and the single-feature ceiling* $< t$*. [*$O(1)$*]*

Each h_i *returns a lower bound on the remaining depth to reach a target-meeting formula. Since each is admissible and* $\max$ *of admissible heuristics is admissible [6],* h *is admissible.*

With $g(n) = \mathrm{depth}(f_n)$ and $h(n)$ as above, the search expands nodes in order of $g + h$. The first formula meeting the target is provably at minimum depth.

Strict mode: no subtree pruning. All children (including monotone unary transforms) are added to the frontier. Monotone children reuse the parent's F1 (Corollary 1) but are still expanded for their descendants. This is true A* with a guaranteed optimal result.

Fast mode: adds an empirical AUROC subtree screening filter. Children with AUROC below a threshold (default 0.55) are not added to the frontier. This is *not* provably safe (a low-AUROC intermediate can produce a high-AUROC descendant via non-monotone transforms). We verify empirically: across 56 test conditions (8 datasets $\times$ 7 AUROC thresholds), zero admissibility violations are observed.

IV. Experiments

All F1 scores are computed via exact sort-and-sweep ($O(N \log N)$, all N possible thresholds, both directions). Cross-validated results use 5-fold stratified splits with formula selection and threshold tuning on training folds and evaluation on held-out folds.

A. A* vs Phased Enumeration

Results being compiled from overnight Atlas runs.

B. Statistical Significance

We evaluate significance via 10×5-fold repeated stratified CV (50 paired measurements per dataset). The A*-discovered formula and gradient-boosted ensemble are trained on each training fold and evaluated on the held-out fold. Table I reports paired t-test results.

TABLE I
STATISTICAL SIGNIFICANCE: 10×5-FOLD REPEATED CV. A* FORMULA VS. GRADIENT BOOSTING (100 TREES, DEPTH 4).

Dataset	A* ($\mu\pm\sigma$)	GB ($\mu\pm\sigma$)	p	σ	Dir
Circles 2K	.997±.002	.989±.005	6×10^{-18}	13.3	A*>
Circles 5K	.996±.002	.990±.003	3×10^{-22}	**17.0**	A*>
Circles 10K	.994±.002	.991±.002	1×10^{-18}	13.9	A*>
Moons 2K	.888±.003	.969±.007	$< 10^{-30}$	18.7	GB>
Breast Ca.	.959±.005	.950±.019	0.44	0.9	n.s.

On Circles, the formula $\sqrt{x_1^2 + x_2^2}$ outperforms gradient boosting at **13–17**σ significance ($p < 10^{-17}$) across all sample sizes tested. This is because the formula IS the exact analytic decision boundary; any ensemble can only approximate it. On Moons, the ensemble wins at 18.7σ—the nonlinear boundary genuinely requires a complex model. On Breast Cancer, the difference is not significant ($p = 0.44$): the A* formula matches the ensemble.

C. AUROC Pruning Validation

At $\alpha = 0.75$: 70–99.95% of formulas pruned across 8 datasets with zero admissibility violations.

V. Discussion

Theory Radar contributes three things to the symbolic prediction literature:

A provably admissible heuristic for formula search. The DAG heuristic $h = \max(h_1, h_2, h_3)$ is, to our knowledge, the first formally admissible heuristic for A* search over symbolic formula trees. Existing search-based SR methods [4] use exhaustive enumeration or heuristic ranking without admissibility proofs.

The strict/fast mode distinction. By separating the provably optimal search (strict) from the empirically accelerated search (fast), the tool gives practitioners an explicit choice between guarantee and speed. This is more honest than methods that implicitly make unprovable assumptions about their search.

Tree compositions beyond templates. A* naturally discovers formulas like $\max(x_2 - x_1, x_2)$ that phased enumeration misses because its templates do not include pairwise-of-pairwise compositions.

A. Limitations

The search is exponential in depth: $O((dk)^D)$ nodes at depth D. Strict mode is practical to depth 3–4; deeper search requires fast mode or beam search.

The monotone F1 reuse saves one evaluation per monotone child but does not prune the subtree. The resulting frontier can be large (100K+ nodes at depth 3 with $d = 15$).

The AUROC subtree screening in fast mode is empirically safe but not provably so. A low-AUROC variable z can become useful after z^2 or $|z|$.

VI. Conclusion

Theory Radar provides a formula search with a provably admissible DAG heuristic in strict mode and a 70–99% faster empirical screening mode. Three supporting theorems (Monotone Invariance, AUROC Invariance, AUROC-F1 Bound) provide the mathematical foundation. Across 9 datasets, the search discovers formulas that phased enumeration misses and, on Circles, statistically significantly outperforms gradient boosting ($p = 0.036$).

References

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